Statistical Thinking for Data Science

Week 2: Turning a vague question into an answerable one

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Lecture notes for students of Statistical Thinking for Data Science.

Week 2: Turning a vague question into an answerable one

Learning Objectives

By the end of this week you should be able to:

- take a vague request and write it as a question that data could settle;
- state the unit of observation for a study, and explain what goes wrong when the unit is chosen carelessly;
- classify a variable by its level of measurement, and say which summaries are then legitimate;
- write an operational definition precise enough that two people measuring separately would record the same number;
- state the estimand: the single quantity your analysis is trying to pin down;
- recognise a proxy, and say in which direction it will mislead you;
- write a data dictionary for a dataset you intend to collect.

2.1 Why the question comes first

Almost nobody will bring you a well-posed question. What arrives is closer to: *"Can you look at the attendance data and see what's going on?"* or *"Management wants to know if the new reagent is working."*

If you start computing at that point you will produce something. It may even be correct. The risk is that it answers a question nobody asked, and the only way to find out is to present it and watch the room go quiet.

Most of the value you add happens before the first line of code, in the conversation where a vague ask becomes a specific question. That conversation is a skill, and like any skill it has a method.

> An analysis cannot be wrong or right until the question is specific. Before that it can only be interesting, which is not the same thing.

2.2 Sharpening a question: five parts

A question is answerable when five things are pinned down.

Part	The question it settles
Units	One row of my data is one _____.
Quantity	I want a number that is a mean, a proportion, a rate, a difference, or a slope.
Comparison	Compared with what? Another group, another period, a target, or nothing.
Context	Which population, which place, which period.
Decision	What would be done differently depending on the answer.

The last one is the one students skip and professionals ask first. If no decision changes whatever the answer, the analysis is for curiosity, and should be budgeted accordingly.

Example 1: from a vague ask to three real questions

"Are students failing PS 361 because they don't attend?"

As stated it cannot be answered: no units, no quantity, no comparison, and two different claims

(about failing and about attending) tangled together. Here are three answerable questions hiding inside it.

1. **Descriptive.** Among students registered for PS 361 in the 2026/2027 first semester (context), what proportion (quantity) attended at least 75% of classes (operational definition), overall and by programme (comparison)? One row per student (unit).
2. **Associational.** Among the same students, what is the difference in mean examination mark between those attending at least 75% of classes and those attending less?
3. **Causal.** If attendance of a student were raised from below 75% to above it, by how much would their expected mark change?

Comment

The three need different data and different care. Question 1 needs only the register. Question 2 needs marks as well, and tells you about an association that may be explained by something else entirely, such as students who work night shifts. Question 3 cannot be answered by the register at all, because the students who attend differ from those who do not in ways that also affect marks. We would need an intervention, or at least a defensible argument about those other differences. Weeks 5 and 11 return to this.

A useful sentence to carry. "So, to be clear: you want the _____ (quantity) of _____ (units) in _____ (context), compared with _____, and you will use it to decide _____." Saying that out loud in the first meeting saves weeks.

2.3 The unit of observation

Definition 2.1 (Unit of observation)**.** The thing that one row of your dataset describes: one student, one blast, one shift, one borehole, one patient-visit, one month.

Getting this wrong is the most common structural error in student projects, and it usually shows up as a dataset where the rows mean two different things at once.

Two rules:

1. **The unit must match the question.** A question about students needs one row per student. If your file has one row per sign-in, you must aggregate first, and decide what to do about a student who signed in twice.
2. **A relationship between aggregates is not the relationship between individuals.** This is important enough to have a name, the *ecological fallacy*, and its own example below.

Example 2: the same data, two opposite conclusions

Six operators, from two shifts, each timed on a standard task.

Shift	Operator	Experience (years)	Minutes per task
A	1	2	30
A	2	3	28
A	3	4	26
B	4	6	40
B	5	7	38
B	6	8	36

Solution

Within shift A, the correlation between experience and time is -1.00: more experienced operators are faster. Within shift B it is also -1.00. Pooling all six rows and ignoring the shift gives $+0.76$: more experienced operators look *slower*.

Nothing is wrong with the arithmetic. Shift B happens to run older equipment (mean 38 minutes against 28), and also happens to hold the experienced operators (mean 7 years against 3). Pooling mixes a within-shift comparison with a between-shift one.
If you reported the pooled figure, the recommendation would be to assign the inexperienced operators to the difficult tasks, which is precisely backwards.

Before you correlate anything, ask what else differs between the groups you have quietly pooled. Week 5 gives this effect its general name, confounding, and Week 11 shows one way of handling it.

2.4 Variables and how they are measured

A variable is a property of a unit that can take different values. The useful classification is by what arithmetic the values support.

Level	Example	What you may legitimately do
Nominal	Programme, ore type	Count, mode, proportions. No ordering.
Ordinal	Likert agreement, severity band	All the above, plus median and percentiles. Means are suspect.
Interval	Temperature in °C, calendar year	Differences are meaningful; ratios are not. 20°C is not twice as hot as 10°C.
Ratio	Mass, time, grade, cost	Everything, including ratios, because zero means none.

Cutting across this: a variable is **discrete** if it counts (number of stoppages) and **continuous** if it measures (duration of a stoppage).

Example 3: the mean of an ordinal scale

Five students rate how often they revise, on the scale 1 = never, 2 = rarely, 3 = sometimes, 4 = often, 5 = always. The answers are $1, 1, 2, 5, 5$.
The mean is 2.8 and the median is 2.

Solution

Reporting "average revision 2.8, between rarely and sometimes" asserts something the scale does not support: that the step from *never* to *rarely* is the same size as the step from *often* to *always*. Nobody has established that.
The median, 2, says only that at least half the group is at *rarely* or below, which is a claim the data genuinely supports. Better still, report the distribution: two *never*, one *rarely*, two *always*, which shows the split that both single numbers hide.

When means are defensible

With many items combined, as in a psychometric scale where seven items are summed, the total behaves closely enough to an interval quantity that means are standard practice. That is an argument from the instrument's validation, not from the labels. We use exactly such a scale later in the course.

2.5 Operational definitions

Definition 2.2 (Operational definition)**.** A rule for measuring a concept, specific enough that two people applying it separately to the same situation would write down the same number.

"Downtime", "attendance", "defect", "poor water quality" all sound precise and none of them is.

Example 4: two honest definitions of downtime, two different plants

A 480-minute shift contains three stoppages: 4, 12 and 45 minutes.

Definition	Stoppages counted	Downtime (min)	Availability
Every stoppage	3	61	87.3%
Only stoppages longer than 10 min	2	57	88.1%

Solution

Availability is $(480-61)/480 = 0.873$ under the first rule and $(480-57)/480 = 0.881$ under the second. A difference of under one percentage point looks harmless, so take the other column seriously: the *number* of stoppages is 3 or 2, a difference of a third. Mean time between stoppages is $480/3 = 160$ minutes or $480/2 = 240$ minutes, which is a 50% difference in a headline reliability figure, from a definition nobody wrote down.

The rule

Fix the definition before you collect, write it in the data dictionary, and never change it mid-study. If you must change it, recompute the whole history on the new definition and say so.

2.6 What counts as an answer: the estimand

Definition 2.3 (Estimand)**.** The single quantity you are trying to learn about, defined precisely enough that someone else could compute it from a complete census of the population.

Naming it prevents the commonest sleight of hand in applied work, which is quietly switching to whichever quantity looks largest.

Example 5: one result, four truthful headlines

A pass rate rises from 20% to 25% after a change in teaching.

Estimand	Value	Reads as
Difference in proportions	5 pp	"Five percentage points better"
Relative change	25%	"A quarter more students passed"
Risk ratio	1.25	"1.25 times the pass rate"
Odds ratio	1.33	"A third higher odds of passing"

Solution

All four are correct. The relative change is $(25-20)/20 = 0.25$, and the odds ratio is

$$\frac{0.25/0.75}{0.20/0.80} = \frac{0.3333}{0.25} = 1.33.$$

A press release will choose 25%. A sceptical reader wants the 5 percentage points, because it says how many students are involved: in a class of 300, fifteen. Report the difference and the baseline together, always.

A trap to remember

"Percent" and "percentage point" are not interchangeable, and the odds ratio is not a risk ratio. The two coincide only when the outcome is rare, which is exactly when people start quoting whichever is bigger.

2.7 Proxies, and the damage they do

Often the thing you care about cannot be measured directly, so you measure something correlated with it: examination marks for understanding, income for poverty, borehole depth for water access, kilometres of road for accessibility. That substitute is a *proxy*.

Proxies are not a sin. Hiding them is. Two rules:

1. **Say which way it is wrong.** A proxy that systematically misses one group biases every comparison involving that group.
2. **Expect it to weaken relationships.** Random measurement error in a variable pulls correlations towards zero, a phenomenon called attenuation.

Example 6: attenuation, and why a weak result may be a measurement problem

Suppose the true correlation between a student's understanding and their later job performance is 0.60, but your measure of understanding is a single test with reliability 0.64. The correlation you can expect to observe is approximately

$$r_{\text{observed}} \approx r_{\text{true}} \sqrt{\text{reliability}} = 0.60 \times \sqrt{0.64} = 0.60 \times 0.80 = 0.48.$$

Solution

Your measurement has destroyed a fifth of the relationship before the analysis begins. Two consequences follow. A weak observed correlation is evidence of either a weak relationship or a noisy instrument, and you cannot tell which from the correlation alone. And improving the measurement, by using several items instead of one, can do more for a study than any amount of modelling afterwards. Week 12 measures reliability properly, on our own data.

2.8 Writing it down: the data dictionary

Everything above is wasted if it lives in your head. A data dictionary is one row per variable, written *before* collection, and it is the single most useful document in an applied project.

Name	Type	Unit	Allowed	Definition
`index_no`	nominal	–	pattern	Student index number, identifier only, never analysed
`programme`	nominal	–	10 codes	Programme of registration this semester
`sleep_hrs`	ordinal	hours	4 bands	Self-reported hours on a normal night this semester
`dass_d`	ratio	points	0–42	Depression subscale: sum of 7 items, doubled
`present`	nominal	–	0/1	1 if signed in before the class code closed

Write the dictionary first and the questionnaire second. Doing it the other way round is how you end up with a dataset containing a column called `other2` that nobody can interpret six months later.

2.9 Eight ways this goes wrong

1. **Definition drift.** The meaning of a variable changes partway through the semester, usually because a new person took over the recording.
2. **Double counting.** A student signs in twice, a blast is logged by two systems, a patient appears under two spellings of their name.

3. **The moving target.** The question is adjusted after the results are seen, so that whatever was found becomes what was asked.

4. **Mixed units.** Half the rows in tonnes, half in kilograms. Always plot the distribution before trusting a column.

5. **Silent missingness.** A blank is recorded as zero, and the average collapses. Decide what a blank means before you compute anything.

6. **Survivorship.** Only the machines still in service are in the maintenance file, so the ones that failed catastrophically are invisible.

7. **Aggregating too early.** Monthly totals arrive and the daily detail is gone, taking with it the only way to answer the question.

8. **Measuring what is easy.** Attendance is recorded because it is simple to record, and then quietly becomes the thing being managed, whether or not it is the thing that matters.

2.10 Summary

Week 2 in six lines

- Pin down units, quantity, comparison, context and the decision before analysing.
- One row of the data must be one of whatever the question is about.
- A relationship between group averages need not hold for individuals.
- The level of measurement decides which summaries are honest.
- Operational definitions decide the numbers; write them down first.
- Name the estimand, and report the difference alongside the baseline.

Exercises

1. Rewrite each as an answerable question, naming units, quantity, comparison and context:
   (a) "Is the new borehole helping?"
   (b) "Are our graduates employable?"
   (c) "Does the night shift have a safety problem?"

2. For each question in Exercise 1, say what decision would change with the answer. If you cannot find one, say so.

3. Give the unit of observation for: a study of fuel use by haul trucks; a study of hostel water supply; a study of examination performance across seven programmes.

4. In Example 2, suppose shift B were given shift A's equipment, so that every time in shift B fell by 10 minutes. Recompute the pooled correlation and comment.

5. Classify by level of measurement: ore grade in g/t; hall of residence; a five-point satisfaction rating; year of manufacture; number of stoppages; temperature in kelvin.

6. A colleague reports "mean satisfaction 3.4 out of 5". Give two reasons to be cautious, and state what you would ask to see instead.

7. Write operational definitions, precise enough for two people to agree, for: (a) a student being present at a lecture; (b) a machine being down; (c) a household having access to safe water.

8. Using Example 4's shift, suppose a fourth stoppage of 9 minutes occurs. Recompute downtime, availability and mean time between stoppages under both definitions, and say which figure changes most in relative terms.

9. A programme raises its pass rate from 45% to 54%. Express the change as percentage points, as a relative change, as a risk ratio and as an odds ratio. Which would you put in a report to the Dean, and why?

10. Your measure of a quantity has reliability 0.81 and the true correlation with an outcome is 0.50. What correlation do you expect to observe? What would the reliability need to be for the observed correlation to reach 0.45?

11. Name a proxy used in your own field. State what it is standing in for, and which group it systematically mismeasures.

12. Write a data dictionary, at least six rows, for a dataset you could collect at UMaT in two weeks. Bring it to class: we will take two of them apart.

Before next week

Read Chapter 1 of *OpenIntro Statistics* on data collection, and bring your data dictionary from Exercise 12. Week 3 asks the next question: granted that you know what you want to measure, who exactly are you going to measure it on, and who will you miss?